

car price predict

April 27, 2026

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[1]: import pandas as pd
import numpy as np
import re
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
import matplotlib.pyplot as plt

url = "https://raw.githubusercontent.com/YBI-Foundation/Dataset/main/
↳Car%20Selling%20Price.csv"
df = pd.read_csv(url)

df = df.drop_duplicates()

def extract_number(x):
    if pd.isna(x):
        return np.nan
    x = str(x)
    num = re.findall(r"\d+\.\d*", x)
    return float(num[0]) if num else np.nan

df['Engine'] = df['Engine'].apply(extract_number)
df['Max_Power'] = df['Max_Power'].apply(extract_number)
df['Mileage'] = df['Mileage'].apply(extract_number)
df['Torque'] = df['Torque'].apply(extract_number)

for col in df.select_dtypes(include=np.number).columns:
    df[col] = df[col].fillna(df[col].median())

for col in df.select_dtypes(include='object').columns:
    df[col] = df[col].fillna(df[col].mode()[0])

df['Car_Age'] = 2024 - df['Year']
df['KM_per_Year'] = df['KM_Driven'] / df['Car_Age']

df = df.drop(['Year', 'Brand'], axis=1)
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df = pd.get_dummies(df, drop_first=True)

X = df.drop('Selling_Price', axis=1)
y = df['Selling_Price']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = RandomForestRegressor(
    n_estimators=200,
    max_depth=15,
    random_state=42,
    n_jobs=-1
)

model.fit(X_train, y_train)

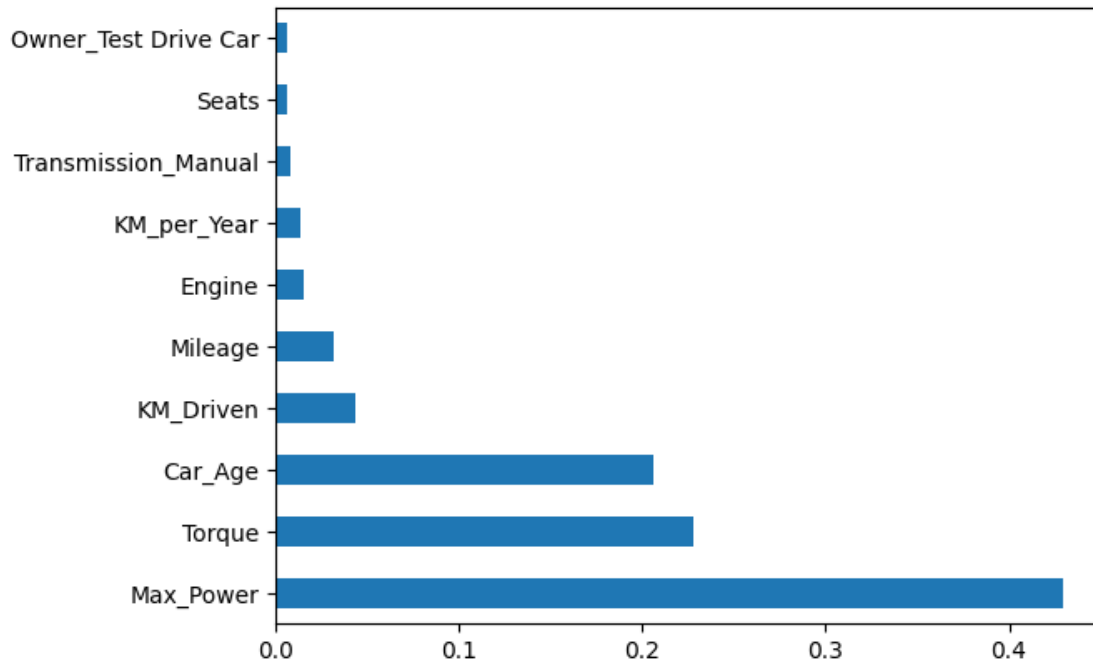
y_pred = model.predict(X_test)

print("MAE:", mean_absolute_error(y_test, y_pred))
print("RMSE:", np.sqrt(mean_squared_error(y_test, y_pred)))
print("R2:", r2_score(y_test, y_pred))

importance = pd.Series(model.feature_importances_, index=X.columns)
importance.sort_values(ascending=False).head(10).plot(kind='barh')
plt.show()

```

MAE: 73463.99791920806
RMSE: 132630.93946010387
R2: 0.9197937464573206



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[2]: import pickle
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pickle.dump(model, open("car_price_model.pkl", "wb"))
pickle.dump(X.columns, open("model_columns.pkl", "wb"))
```

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[3]: sample = pd.DataFrame([
    "Year": 2015,
    "KM_Driven": 50000,
    "Fuel": "Diesel",
    "Transmission": "Manual",
    "Owner": "First Owner"
])
```

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[4]: # Feature engineering
sample['Car_Age'] = 2024 - sample['Year']
sample['KM_per_Year'] = sample['KM_Driven'] / sample['Car_Age']

sample = sample.drop(['Year'], axis=1)

# Encoding
sample = pd.get_dummies(sample)

# Match training columns
sample = sample.reindex(columns=X.columns, fill_value=0)
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[5]: prediction = model.predict(sample)
      print("Predicted Price:", prediction[0])
```

Predicted Price: 152724.4464102564